FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – I

JEE (Advanced)-2021

PAPER –1

TEST DATE: 06-06-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. $KE_{rod} = \frac{1}{2}\mathrm{I}_{\mathrm{cm}}\omega^2 + \frac{1}{2}\frac{1}{2}mv_{cm}^2 = \frac{1}{2}\left(\frac{M\ell^2}{12}\right)\frac{v_0^2}{\ell^2} + \frac{1}{2}M\left(\frac{3}{2}v_0\right)^2$

2. C

Sol. $f \propto \sqrt{T}$

So Δf increases by increasing T.

i. e. $f_2 = f_1 + 3 = 443Hz$

3. C

Sol. $\bar{u}_{cm} = \frac{m\bar{u}_1 + 2m\bar{u}_2}{3m} = \frac{\bar{u}_1 + 2\bar{u}_2}{3}$

$\bar{a}_{cm} = \frac{m\bar{a}_1 + 2m\bar{a}_2}{3m} = \frac{\bar{a}_1 + 2\bar{a}_2}{3}$

4. B

Sol. If two uniform spherical drops of a liquid

$\frac{4}{3}\pi R^3 = 2\left(\frac{4}{3}\pi r^3\right) \Rightarrow R = 2^{1/3}r$

$$\Rightarrow v_2 = \frac{2rv_1}{R} = \frac{2r(16)^{1/3}}{(2)^{1/3} r} = 4\,m/s$$

5. C

Sol. $\frac{V_{gas}}{V_{air}} = \frac{f\lambda_{gas}}{f\lambda_{air}} = \frac{\Delta\ell'}{\Delta\ell}$

$$V_{gas} = \frac{1000}{3} \times \frac{3}{2}\frac{\Delta\ell}{\Delta\ell} = 500\,ms^{-1}$$

6. C

Sol. $1\sin 45^0 = \sqrt{2}\sin\theta$

$\theta = 30^0$

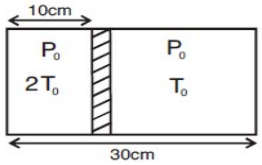
At first refraction

$\delta_1 = 45^0 - 30^0 = 15^0$

And at second refraction

$\delta_2 = 45^0 - 30^0 = 15^0$

Thus $\delta_{net} = 15^0 + 15^0 = 30^0$



7. BC

Sol. FBD of rope in contact with pulley is shown here we can see torque of dN about O is zero Torque of friction is balanced by torque T_1 and T_2

8. BD

Sol. $W_g + W_{fr} = \Delta K$

W_g & ΔK same for all cases so W_{fr} should also be same

$N_1 < N_2$ & $\ell_1 < \ell_2 \Rightarrow \mu_1 > \mu_2$ $\quad (N = normal\ force, \ell = length\ of\ the\ path)$

$N_2 > N_3$ & $\ell_2 = \ell_3 \Rightarrow \mu_2 < \mu_3$

9. BCD

Sol. μ required for pure rolling

Therefore $f = \mu mg\cos\theta = \frac{mg\sin\theta}{4}$

$a_{cm} = 3g\frac{\sin\theta}{4}$, time to each bottom $t = \sqrt{\frac{8h}{3g\sin^2\theta}}$

$$a = \frac{mg\sin\theta}{4mR^2}R = \frac{g\sin\theta}{4R}$$

$$\therefore \theta_{rotated} = \frac{1}{2} \times \frac{g\sin\theta}{4R} \times \frac{8h}{3g\sin^2\theta} = \frac{h}{3R\sin\theta}$$

KE at bottom $= \frac{5mgh}{6}$

10. AC

Sol. Two particles 1 and 2 are moving

Acceleration of 2 with respect to 1 is $\vec{a}_2 - \vec{a}_1$

Angular velocity of 2 with respect to 1 is same as angular velocity of 1 with respect to 2

11. CD

Sol. $h = \sqrt{9 \times 4} = 6cm$

$$\left.\begin{aligned} \frac{v}{u} &= \frac{9}{h} \\ \frac{u}{v} &= \frac{4}{h} \end{aligned}\right\} \Rightarrow \frac{v}{u} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

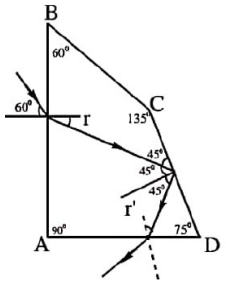
$$u + v = 90 \Rightarrow u + \frac{3}{2}u = \frac{5}{2}u = 90 \Rightarrow 36cm$$

12. ABD

Sol. $\frac{\sin 60}{\sin r} = \sqrt{3} \Rightarrow \sin r = \frac{1}{2} \Rightarrow r = 30° \sin c = \frac{1}{\sqrt{3}} \Rightarrow c = 35.3°$

$45° > C \Rightarrow$ total internal reflection at face CD.

$r' = 90 - (180 - 45 - 75) = 120 - 90 = 30°$



SECTION – C

13. 00002.00

Sol. Initial charge (q_i) on the capacitor $= 36V \times 250\,mF = 9C$

Final charge (q_f) on the capacitor $= \left[\frac{4}{12+4} \times 12\right] \times \frac{250}{1000} C = \frac{3}{4} C.$

Time constant (t) of the circuit $= \left(\frac{4 \times 12}{4+12} + 3\right) \times \frac{250}{1000} s = 1.5\,s.$

Equation of charge on the capacitor $= q_f + (q_i - q_f) e^{\frac{i}{t}} = \frac{3}{4} + \left(9 - \frac{3}{4}\right) e^{\frac{-2t}{3}} = \frac{3}{4} + \frac{33}{4} e^{\frac{-2t}{3}}$

Equation of current in the $3\ \Omega$ resistor $i = \frac{-dq}{dt} = \frac{11}{2} e^{\frac{-2t}{3}} A.$

Current at $t = 3$ ln 2s $i = \frac{11}{2} e^{-\frac{2}{3} \times 3 In} = \frac{11}{8} A.$

$So\ x = 2$

14. 00008.00

Sol. $Q = 4 \times 7.1 - 2 \times 1.1 - 2 \times 1.1 = 24\text{MeV}$

So, $\frac{Q}{3} = 8\text{MeV}$

15. 00300.00

16. 00525.00

17. 00009.00

Sol. ice at 0°C is added to 200 gm

$$C(70-40)+200\times(70-40)\times1 = 50L+50\times1\times(40-0)$$

$$C(40-10)+250(40-10)\times1 = 80L+80\times1\times(10-0)$$

$$L=90\,\text{cal/gm}$$

18. 00004.00

Sol. $$T=2\pi\sqrt{\frac{I}{C}}=2\pi\sqrt{\frac{2}{5}\frac{mR^2}{C}}$$

$$I=\frac{2}{5}mR^2$$

Chemistry

PART – II

SECTION – A

19. A

20. C

21. B

22. C

Sol. $r = k[A][B_2]$ & $k_c = \dfrac{[A][A]}{[A_2]}$

$\therefore [A] = K^{1/2}[A_2]^{1/2}$

Thus $r = K.K_c^{1/2}[A_2]^{1/2}[B_2] = K'[A_2]^{1/2}[B_2]$

Hence order reaction s 3/2

23. B

24. A

Sol. $BaO_2.8H_2O + H_2SO_4 \longrightarrow BaSO_4 \downarrow + H_2O_2 + 8H_2O$

$H_2O_2 + H_2SO_4 + 2KI \longrightarrow K_2SO_4 + 2H_2O + I_2$

25. ACD

Sol. (A) As reaction is endothermic therefore it will go in the forward direction hence moles of CaO will increase.

(C) With the increase or decrease of volume particle pressure of the gases will remain same.

(D) Due to the addition of inset gas at constant pressure reaction will proceeded in the direct in which more number of gaseous are formed.

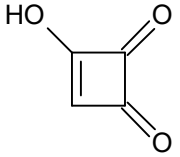
26. ACD

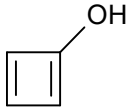
Sol. $W_{\text{on the system, adiabatic}} > W_{\text{on the system, isothermally}}$

$MgCO_3(s) \longrightarrow MgO(s) + CO_2(g)$

27. ABD

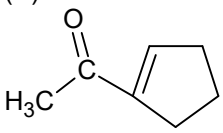
Sol. $A \rightarrow$ hydrogen of diketone is more acidic.

$B \rightarrow$  has some aromatic character

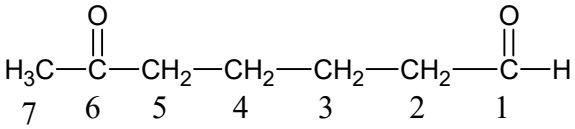
$C \rightarrow$  is antiaromatic

28. AB

Sol. (A)



Aldol



Here C-2 and C-5 can form carbanion & leads to different product.

29. BCD

Sol. Aniline can show Carbylamine test, Diazo test, while Benzamide shows reaction with $NaOH+Br_2$.

30. BCD

Sol. Lyophilic colloids are used to protect lyophobic colloid.

SECTION – C

31. 00000.09

Sol. $\underset{1}{PCl_5} \rightleftharpoons \underset{0}{PCl_3} + \underset{0}{Cl_2}$

$1-\alpha \qquad \alpha \qquad \alpha$

$1+\alpha = \frac{D}{d}$

$1+\alpha = \frac{250.2}{208.5}$

$\alpha = 0.2$

Mole of $PCl_5 = 0.8$, PCl_3 $Cl_2 = 0.2$

$\therefore K = \frac{0.2\times 0.2}{0.8} = \frac{1}{20}$

Now, when it transferred to 2L vessel reaction proceeds in forward direction.

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$0.8-x \qquad 0.2+x \quad 0.2+x$

$\frac{(0.2+x)^2}{(0.8-x)\times 2} = \frac{1}{20}$

$\therefore x = 0.07$

Degree of dissociation $= \frac{0.07}{0.08} = 0.0875(\text{approx}) = 0.09$

32. 00021.00

Sol. According to the law of conservation of momentum, momentum lost by photons
= momentum gained by glow-worm
Momentum lost by photons
= no of photons x momentum lost by one photon

$= \frac{\text{Total energy}}{\text{energy of 1photon}} \times \frac{h}{\lambda}$ $\left[\text{Debroglie's equation } \lambda = \frac{h}{p}.p = \frac{h}{\lambda}\right]$

$$= \frac{\text{power} \times \text{time}}{\frac{hc}{\lambda}} \times \frac{h}{\lambda}$$

$$= \frac{\lambda}{hc} \times \text{power} \times \text{time} \times \frac{h}{\lambda}$$

$$= \frac{0.1 \times 365 \times 24 \times 60 \times 60 \times 10}{3 \times 10^8} \Rightarrow (1)$$

Momentum gained by glow-worm = $\frac{5}{1000} \times (v) \text{ in m/sec} \longrightarrow (2)$

(1) = (2)

Hence v = 21 m/sec

Ans. $\frac{v}{7} = 3$.

33. 00006.90

Sol. $10Al + 6KMnO_4 + 27H_2SO_4 \longrightarrow 6KHSO_4 + 5Al_2(SO_4)_3 + 6MnSO_4 + 24H_2O$

34. 00001.75

Sol: H_2SO_4 produced = $\frac{2.463 \times 0.01}{2 \times 0.0821 \times 300} = \frac{10^{-2}}{2}$M

$2NO_2 + H_2O \rightleftharpoons HNO_2 + HNO_3$

Conc. of $NO_2 = \frac{10^{-2}}{2}$M

So $HNO_2 = \frac{10^{-2}}{2}$M

$HNO_3 = \frac{10^{-2}}{2}$M

Net $\left[H^+\right] = \frac{10^{-2}}{2} \times 2 + \frac{10^{-2}}{2} + \frac{10^{-2}}{2} \times \frac{60}{100} = 1.8 \times 10^{-2}$

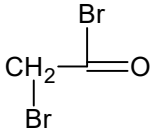
pH = 1.75

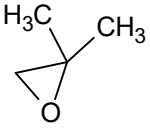
35. 00357.90

Sol. One mole of compound consume 6 moles of CH_3MgBr so half mole will consume 3 moles of CH_3MgBr

36. 00072.00

Sol.

A is 

and B is 

Mathematics

PART – III

SECTION – A

37. B

38. B

Sol. Use the concept of weighted AM. And G.M

39. C

Sol. Parabola is $(y + 1)^2 = 8 (x – 4)$

Focus = (2, 0)

Directrix: x = 2.

Clearly P (2, α) lies on the directrix of the parabola and $x + \beta y – 2 = 0$ is the equation of any focal chord.

40. B

Sol. $f(x) - f(x/2) = x/2$

$f(x/2) - f(x/4) = x/4$

$f(x/4) - f(x/8) = x/8$

$f\left(\frac{x}{2^{n-1}}\right) - f\left(\frac{x}{2^{xn}}\right) = \frac{x}{2^{xn}}$

$\therefore \lim_{x\to\infty}\left(f(x) - f\left(\frac{x}{2^n}\right)\right) = x$

41. C

Sol. Suppose f(x) has degree < 2 $\Rightarrow$ f(x) = bx + c

$\Rightarrow$ bx + c + 1 = bx + c not possible

suppose f(x) has degree > 2 $\Rightarrow f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots\ldots \quad (a \neq 0)$

comparing coefficients of x^{n-2} we get

$c = \frac{a(n)(n-1) + 2c}{2} \Rightarrow a = 0.$

$\Rightarrow$ f(x) has degree 2.

42. B

Sol. Given equation is

$\cos x . \sin x – \cos^2 x = \frac{n}{2}$.

$\Rightarrow \frac{1}{2}\sin 2x - \left[\frac{1+\cos 2x}{2}\right] = \frac{n}{2}$

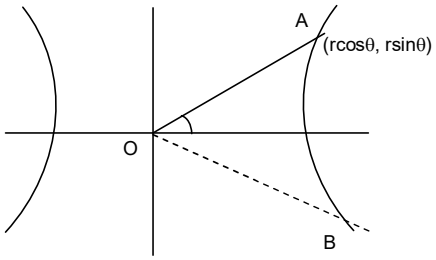
$\Rightarrow \sin 2x + \cos 2x – 1 = n$

$\Rightarrow -\sqrt{2} \le n + 1 \le \sqrt{2}$

$\Rightarrow$ Greatest integral value of n is '0'

43. AC

Sol. Sub (r cosθ, r sinθ) in the equation of hyperbola



$$\frac{1}{r^2}=\frac{\cos^2\theta}{a^2}-\frac{\sin^2\theta}{b^2}$$

$$\frac{1}{r_1^2}-\frac{1}{r_2^2}=\frac{\cos^2\theta_1-\cos^2\theta_2}{a^2}-\frac{\sin^2\theta_1-\sin^2\theta_2}{b^2}$$

$\theta_2 = \theta_1 + \frac{\pi}{2}$

$$\therefore\ \frac{1}{r_1^2}-\frac{1}{r_2^2}=\frac{\cos^2\theta_1-\sin^2\theta_1}{a^2}-\frac{\sin^2\theta_1-\cos^2\theta_1}{b^2}$$

$$=\frac{\cos^2\theta_1-\sin^2\theta_1}{a^2}-\frac{\sin^2\theta_1-\cos^2\theta_1}{b^2}=(\cos^2\theta_1-\sin^2\theta_1)\left(\frac{1}{a^2}+\frac{1}{b^2}\right)$$

$$\max\left|\frac{1}{r_1^2}+\frac{1}{r_2^2}\right|=\frac{1}{a^2}+\frac{1}{b^2}$$

$$\frac{1}{r_1^2}+\frac{1}{r_2^2}=\frac{\cos^2\theta_1}{a^2}-\frac{\sin^2\theta_1}{b^2}+\frac{\cos^2\theta_2}{a^2}-\frac{\sin^2\theta_2}{b^2}=\frac{\cos^2\theta_1+\sin^2\theta_1}{a^2}-\frac{\sin^2\theta_1+\cos^2\theta_1}{b^2}=\frac{1}{a^2}-\frac{1}{b^2}$$

$\frac{1}{r_1^2}+\frac{1}{r_2^2}$ is constant.

44. ABCD

Sol. (A) $\frac{a^2}{1}+\frac{b^2}{1}\ge\frac{(a+b)^2}{2}$

$$\Rightarrow \frac{a^2+b^2}{a+b}\ge\frac{a+b}{2}$$

$$\Rightarrow \sum\frac{a^2+b^2}{a+b}\ge a+b+c=8$$

(C) as $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\ge\frac{1}{ab}+\frac{1}{bc}+\frac{1}{ca}$

$a^2b^2 + b^2c^2 + c^2a^2 \ge abc(a + b + c)$

(D) $a^2 + b^2 + c^2 \ge ab + bc + ca$

$3(a^2 + b^2 + c^2) \ge (a + b + c)^2$

$= 64$

45. AC

Sol. $\sin\beta + \sin\gamma + \sin\delta = 1 - \sin\alpha$

$\cos2\alpha + 1 - 2\sin^2\beta + 1 - 2\sin^2\gamma + 1 - 2\sin^2\delta \ge 10/3$

$$\Rightarrow \cos2\alpha\ge\frac{1}{3}+2\left(\sin^2\gamma+\sin^2\beta+\sin^2\delta\right)$$

$$\Rightarrow 1-2\sin^2\alpha\ge\frac{1}{3}+2\frac{(\sin\gamma+\sin\beta+\sin\delta)^2}{3}=\frac{1+2(1-\sin\alpha)^2}{3}$$

$\Rightarrow 3 - 6\sin^2\alpha \ge 2\sin^2\alpha - 4\sin\alpha + 3.$

$$\Rightarrow 8\sin^2\alpha-4\sin\alpha\le0\Rightarrow\alpha\in\left[0,\frac{\pi}{6}\right]$$

$\therefore$ A,C.

46. CD

47. AC

Sol. $\sin\sqrt{2} - \sin\sqrt{3} = 2\sin\frac{\sqrt{2}-\sqrt{3}}{2}\cos\frac{\sqrt{2}+\sqrt{3}}{2}$

$\frac{\sqrt{2}-\sqrt{3}}{2} < 0,\ \frac{\sqrt{2}+\sqrt{3}}{2} > \frac{\pi}{2}$

$\Rightarrow \cos\frac{\sqrt{2}+\sqrt{3}}{2} < \cos\frac{\pi}{2}$

$\cos\frac{\sqrt{2}+\sqrt{3}}{2} < 0$

$\Rightarrow \sin\sqrt{2} - \sin\sqrt{3} > 0$

$\cos\sqrt{2} - \cos\sqrt{3} > 0$ (since $\cos\sqrt{2} > 0, \cos\sqrt{3} < 0$)

$\Rightarrow$ the equation is an ellipse.

Consider $\sin\sqrt{2} - \sin\sqrt{3} - \cos\sqrt{2} + \cos\sqrt{3}$

$= (\sin\sqrt{2} - \cos\sqrt{2}) - (\sin\sqrt{3} - \cos\sqrt{3})$

$= \sqrt{2}\left(\frac{1}{\sqrt{2}}(\sin\sqrt{2} - \cos\sqrt{2}) - \frac{1}{\sqrt{2}}(\sin\sqrt{3} - \cos\sqrt{3})\right)$

$= \sqrt{2}\left(\sin\left(\sqrt{2} - \frac{\pi}{4}\right) - \sin\left(\sqrt{3} - \frac{\pi}{4}\right)\right)$

Both $\sqrt{2} - \frac{\pi}{4} < \frac{\pi}{2}, \sqrt{3} - \frac{\pi}{4} < \frac{\pi}{2}$

And $\sqrt{3} - \frac{\pi}{4} > \sqrt{2} - \frac{\pi}{4}$

$\Rightarrow \sin\sqrt{2} - \sin\sqrt{3} - \cos\sqrt{2} + \cos\sqrt{3} < 0$

$\Rightarrow$ The major axis is y–axis

48. AB

Sol. Let (l, m, n) be the direction cosines of the line perpendicular to the plane.

$\Rightarrow$ Equation of the plane $lx + my + nz = p$

$\frac{x}{\left(\frac{p}{l}\right)} + \frac{y}{\left(\frac{p}{m}\right)} + \frac{z}{\left(\frac{p}{n}\right)} = 1$

$A(p/l,0,0), B(0,p/m,0), C(0,0,p/n)$

centroid of tetrahedron OABC is

$(x, y, z) = \left(\frac{p}{4l}, \frac{p}{4m}, \frac{p}{4n}\right)$

Using $l^2 + m^2 + n^2 = 1$

$x^2y^2 + y^2z^2 + z^2x^2 = \frac{16}{p^2}x^2y^2z^2$

$\Rightarrow \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}$

Put $x = \frac{p}{4}\sec\alpha\sec\beta$, $y = \frac{p}{4}\sec\alpha\,\mathrm{cosec}\beta$, $z = \frac{p}{4}\mathrm{cosec}\alpha$

$\frac{1}{x} = \frac{4}{p}\cos\alpha\cos\beta, \frac{1}{y} = \frac{4}{p}\cos\alpha\sin\beta, \frac{1}{z} = \frac{4}{p}\sin\alpha$

$\left(\frac{1}{x}\right)^2+\left(\frac{1}{y}\right)^2+\left(\frac{1}{z}\right)^2=\frac{16}{p^2}\left[\cos^2\alpha\cos^2\beta+\cos^2\alpha\sin^2\beta+\sin^2\alpha\right]$

$=\frac{16}{p^2}\left[\cos^2\alpha+\sin^2\alpha\right]=\frac{16}{p^2}$

SECTION – C

49. 00017.00

50. 00002.00

Sol. $|z_1+1|+|z_2+1|+|z_1z_2+1|$

$\geq |z_1+1|+|z_2+1-z_1z_2-1|$

$\geq |z_1+1|+|z_2||z_1-1|=|z_1+1|+|z_1-1|$

$|z_1+1|+|z_1-1|\geq 2$

51. 00001.97

Sol. $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2=140$

$\frac{1}{6}\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\frac{11.83}{6}=1.97$

52. 00002.40

Sol. $|z-1|+|z-2|+|z-3|\leq|3z-6|=3|z-2|=3|2e^{i\theta}-2|=12$

53. 00000.25

Sol. All points lies on circle $x^2+y^2-\frac{y}{2}=0$

$\therefore$ radius $=\frac{1}{4}$

Area of regular octagon $=8\left(\frac{1}{2}\times r^2\times\sin\left(\frac{2\pi}{8}\right)\right)$

$$A=\frac{1}{4\sqrt{2}}$$

54. 00004.27

Sol. $A\equiv\left(\frac{16}{5}\cos\theta,0\right)$, $B\equiv\left(0,\frac{-16}{3}\sin\theta\right)$

Area of $\Delta OAB=\frac{1}{2}\left|\frac{16}{5}\cos\theta\right|\left|\frac{-16}{3}\sin\theta\right|=\frac{1}{4}.\frac{16.16}{15}|\sin2\theta|$

$\leq\frac{64}{15}=4.27$